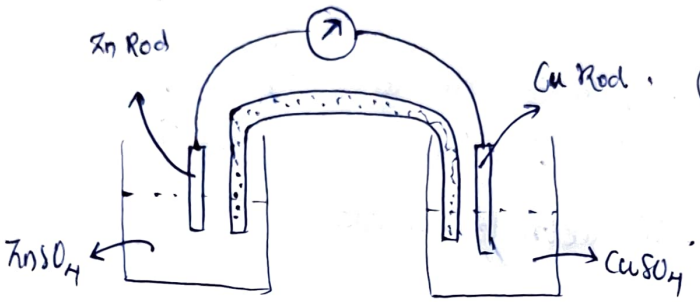
ELECTROCHEMISTRY.

chemical ⟶ Electrical energy. (cell)

Galvanic cell ⟶ Daniel Cell.

Anode ⟶ oxidⁿ ⟶ electron loss.

Cathode ⟶ redⁿ ⟶ Gain of $\bar{e}$.



Daniell Cell

1.1V

$Zn \longrightarrow Zn^{2+} + 2\bar{e}$

$Cu^{2+} + 2\bar{e} \longrightarrow Cu$

$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$

125 DAY.

* $E_{cell} = E_{cathode} - E_{anode}$

* $E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$

↓ → Standard - fixed.

at 1atm, 1M, 298K

~~* Electrode potential~~

$E^{\circ}_{cell} = E^{\circ}_{right} - E^{\circ}_{left}$

⇒ Electrode Potential.

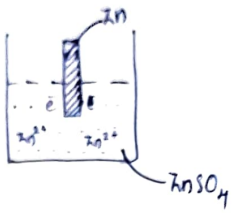
- Tendency of an electrode either to get oxidised or reduced.
- if oxidised ; oxidation potential. $E^{\circ}_{Zn} = +0.76V$ $Zn \longrightarrow Zn^{2+} + 2e$
- if reduced ; reduction potential. $E^{\circ}_{Zn} = -0.76V$ $Zn^{2+} + 2\bar{e} \longrightarrow Zn$

* By convention, reduction potential is taken as Electrode potential

eg. $E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$

Q: How does single electrode potential arise?

due to separation of charges b/w electrode & soln, an electric potential is set up b/w the metal electrode and its soln



Electrode potential ; The potential difference b/w the metal electrode and the solution containing its ions surrounding the electrode is called its electrode potential.

Q: How to measure ....

$E_{cell} = E_{unknown} - E_{known}$

By connecting it with a known potential as reference

* Reference electrodes ⟹ known, the potential will be constant for them.

most commonly used ; "Standard Hydrogen Electrode (SHE)"

→ SHE.

* condition ; 1atm, H^+, 1M, 298K.

* The electrode potential of SHE is fixed as 0V for convenience.

→ Electrochemical Series. (Rising order of E°_{cell})

red^n potential

oxidation potential of Reactant

| K, Na
| Ca
| Mg
| Al

Zn
Fe
Ni
Sn
Pb
H
Cu
Hg
Ag
Pt
Au

E_{red^n} ↑ as we go down

★ the metal above displaces the metal below.

⇒ Fe - Cu Cell ;

Fe → anode → oxidn

Cu → cathode → redn

Cell Notation

$Fe \mid Fe^{2+}_{(xM)} \parallel Cu^{2+}_{(yM)} \mid Cu$

Fe | $Fe^{2+}_{(xM)}$: Anode

$Cu^{2+}_{(yM)}$ | Cu : cathode

$Fe \rightarrow Fe^{2+} + 2e^-$

$Cu^{2+} + 2e^- \rightarrow Cu$

$Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$ electrode R^n.

⇒ Zn - H_2 Cell ;

Zn → Anode → Oxidn

H_2 → Cathode → Redn

CN ;

$Zn \mid Zn^{2+}_{(xM)} \parallel H^+_{(1M)} \mid H_2, Pt_{(1atm)}$

$Zn \rightarrow Zn^{2+} + 2e$

$2H^+ + 2\bar{e} \rightarrow 2H_2$

$Zn + 2H^+ \rightarrow Zn^{2+} + H_2$

⇒ H_2 - Cu Cell

↓ Anode (oxdn) ↓ Cathode (redn)

∴ Cell notatn ;

$2H_2 \rightarrow 2H^+ + 2e^-$ $\quad Pt, H_2 \mid H^+ \parallel$ ✗

$Cu^{2+} + 2\bar{e} \rightarrow Cu$

$H_2 + Cu^{2+} \rightarrow 2H^+ + Cu$

$H^+ \mid H_2, Pt$ ✓

↙ when Cathode

∴ $H^+/H_2, Pt \,//\, Cu^{2+}/Cu$ (1atm)

Q: Write the cell-notation for ;

① Mg - Fe Cell.

Mg → anode → oxdn ⇒ $Mg \rightarrow Mg^{2+} + 2e^-$

Fe → Cathode → redn ⇒ $Fe^{2+} + 2e^- \rightarrow Fe$

Cell rxn ; $Mg + Fe^{2+} \rightarrow Mg^{2+} + Fe$

Cell notn ; $Mg \mid Mg^{2+}_{(1M)} \parallel Fe^{2+}_{(1M)} \mid Fe$

② Ag – Ni Cell

Ag → anode ⇒ oxidn → $2Ag \longrightarrow 2Ag^{+} + 2e^{-}$

Ni ⇒ cathode → redn ⇒ $Ni^{2+} + 2e^{-} \longrightarrow Ni$

cell rxn ; $2Ag + Ni^{2+} \longrightarrow 2Ag^{+} + Ni$

Cell notn ; $Ag \mid \underset{(xM)}{Ag^{+}} \parallel \underset{(yM)}{Ni^{2+}} \mid Ni$

③ Ni – H Cell

Ni → anode → oxidn ⇒ $Ni \longrightarrow Ni^{2+} + 2e^{-}$

H → Cathode → redn ⇒ $2H^{+} + 2e^{-} \longrightarrow H_2$

Cell rxn ; $Ni + 2H^{+} \longrightarrow Ni^{2+} + H_2$

Cell notn ; $Ni \mid \underset{(xM)}{Ni^{2+}} \parallel \underset{(yM)}{H^{+}} \mid \underset{(1atm)}{H_2, Pt}$

11/25
esday.

* $H_2 \longrightarrow 2H^{+} + 2e^{-}$ (oxidn)
* $2H^{+} + 2e^{-} \longrightarrow H_2$ (g) (redn)

ie ; If H_2 gas is to be evolved ; H_2 needs to be in redn

⇒ NERNST EQUATION.

Q → reactn constant.
↳ $\frac{[P]}{[R]}$

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

$$\Delta G = -nFE^{\circ}_{cell}$$

$$-nF E_{cell} = -nF E^{\circ}_{cell} + RT \ln \frac{[Pdt]}{[Rct]}$$

dividing by $-nF$

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF} \ln \frac{[Pdt]}{[Rct]}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{2.303\,RT}{nF} \log \frac{[Pdt]}{[Rct]}$$

$$\boxed{E_{cell} = E^{\circ}_{cell} + \frac{0.0591}{n} \log \frac{[Rct]}{[Pdt]}}$$

For Daniel Cell ;

$$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$$

$$E_{cell} = E^{\circ}_{cell} + \frac{0.0591}{2} \log \frac{[Cu^{2+}]}{[Zn^{2+}]}$$

$$Zn^{2+} + 2e^{-} \longrightarrow Zn \qquad = -0.76\,V$$

$$= -0.76 + \frac{0.0591}{2} \log \frac{(3.5)}{1}$$

Q: For the clanicll cell ; the standard free energy is $-218.4\,KJ$ Calculate the Std. EMF

$$\Delta G^{\circ} = -218.4\,KJ$$

$$E^{\circ}_{cell} = ?$$

$\Delta G^\circ = -nF E^\circ_{cell}$

$E^\circ_{cell} = \dfrac{\Delta G^\circ}{-nF}$

$= \dfrac{-218.4 \times 10^3 \text{ J}}{-2 \times 96500 \text{ C}}$

$= 1.13 \text{ V}$

$= 1.1 \text{ V}$

Q: Calculate the emf of the cell $Fe | Fe^{2+}_{(1M)} // Ni^{2+}_{(1M)} | Ni$ at 298K

given, $E^\circ_{Fe} = -0.44 \text{ V}$

$E^\circ_{Ni} = +0.25 \text{ V}$

Soln

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[Red] \to 1}{[ox] \to 1}$$

$\log 1 = 0$

$Fe \longrightarrow Fe^{2+} + 2\bar{e}$

$Ni^{2+} + 2\bar{e} \longrightarrow Ni$ $\quad n = 2$

$Fe + Ni^{2+} \longrightarrow Fe^{2+} + Ni$

$E^\circ_{cell} = E^\circ_{Ni} - E^\circ_{Fe} = 0.25 - -0.44 = 0.69 \text{ V}$

$E_{cell} = 0.69 - 0$

$E_{cell} = 0.69 \text{ V}$

Q. Calculate the emf of cell ;

$Zn \,|\, Zn^{2+} \,//\, Ag^{+} \,/\, Ag$ at 298 K.

(0.001M) (0.1M)

$E^{\circ}_{Zn} = -0.76\ V$

$E^{\circ}_{Ag} = +0.8\ V$

$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$

$= 0.8 - -0.76$

$= 1.56\ V$

$Zn + 2Ag^{+} \longrightarrow 2Ag + Zn^{2+}$ $\quad n = 2$

$$E_{cell} = 1.56 + 0.02955 \log \frac{[Ag^{+}]^2}{[Zn^{2+}]}$$

$$= 1.56 + 0.02955 \log \frac{(0.001)^2}{(0.1)^2} \quad \frac{(0.1)^2}{0.001}$$

$$= 1.56 + 0.02955 \log \frac{(10^{-3})^2}{10^{-1}} \quad \frac{10^{-2}}{10^{-3}}$$

$$= 1.56 + 0.02955 \log \frac{10}{10}$$

$= 1.56 -$

$= 1.53045$

$= 1.58955$

Electrolysis

* electrolyte is broken down into its ions.
* electrical energy ⟶ chemical energy.

$Na^+ - Cl^-$ Electrolytic Cell.

At ~~Cat~~ Anode ;

$$2Cl^- \longrightarrow Cl_2 + 2e^-$$

(Cl_2 : liberated)

At Cathode ;

$$2Na^+ + 2e^- \longrightarrow 2Na$$

(Na : deposited)

$$2Na^+ + 2Cl^- \longrightarrow \uparrow Cl_2 + 2Na$$

→ Faraday's First law.

* The amnt of substance liberated or deposited at electrode during electrodeg is directly proportional to the amnt of charge passing through it.

$$m \propto Q$$

$$m = ZQ$$

$$m = ZIt$$

$1Ag \rightarrow 1e^-$

$1e^- \simeq 96500\,C \Rightarrow 1F$

$Cu^{2+} \rightarrow 2e^-$

$2e^- \rightarrow 2 \times 96500\,C \Rightarrow 2F$

mass of A

→ Faraday's second law.

Q:- Calculate the quantity of electricity required to deposit 6.354 g Cu. Eq mass = 63.54 g

$63.54\ g\ Cu \equiv 2F$

$6.354\ g\ Cu \equiv$

$\therefore 1F \equiv \frac{63.54}{2}$

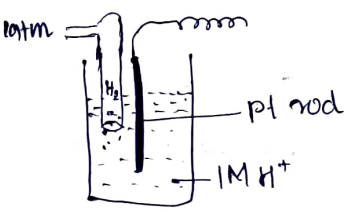
$1g \equiv \frac{2}{63.54} F$

$6.354\ g = \frac{2 \times 6.354}{63.54} \cdot F$

$= 0.2\ F$

$= 19300\ C$

→ Reference Electrodes. → SHE



298K

• if connected with Zn electrode ;

$Zn \rightarrow$ Anode

$Zn \longrightarrow Zn^{2+} + 2\bar{e}$

$SHE \rightarrow$ Cathode

$2H^+ + 2\bar{e} \longrightarrow H_2$

$Zn / Zn^{2+} \;//\; H^+ / H_2, Pt$

(1M) (1atm)

if connected with Cu electrode ;

$Cu \longrightarrow$ Cathode

$Cu^{2+} + 2\bar{e} \longrightarrow Cu$

$SHE \rightarrow$ Anode

$H_2 \longrightarrow 2H^+ + 2\bar{e}$

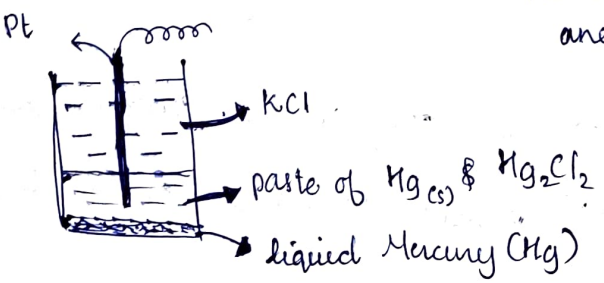
$H_2, Pt / H^+ \;//\; Cu^{2+} / Cu$

(1atm) (1M)

* Secondary Reference Electrodes.

1. Calomel Electrode

* can either be cathode or anode



* Hg_2Cl_2 (mercurous chloride) is a sparingly soluble soln.

As Cathode ;

$Hg_2Cl_2 \longrightarrow Hg^{2+} + 2Cl^-$ M

$Hg_2^{2+} + 2e^- \longrightarrow 2Hg$

$Hg_2Cl_2 + 2e^- \longrightarrow 2Hg + \underset{\text{uuu}}{2Cl^-}$ ★ chloride ions liberated.

$$E_{cal} = E^\circ_{cal} + \frac{0.0591}{2} \log \frac{[Hg_2Cl_2]}{[Hg][Cl^-]^2}$$

$$E_{cal} = E^\circ_{cal} + -0.0591 \log [Cl^-]$$

As Anode ;

$2Hg \longrightarrow Hg_2^{2+} + 2e^-$

$Hg^{2+} + 2Cl^- \longrightarrow Hg_2Cl_2$

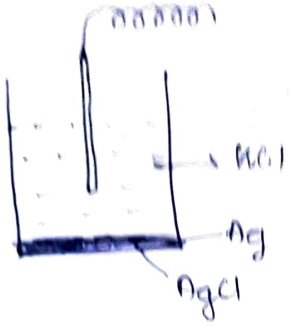
$2Hg + Hg^{2+} + 2Cl^- \longrightarrow Hg^{2+} + Hg_2Cl_2$

$2Hg + \underset{\text{uuu}}{2Cl^-} \longrightarrow Hg_2Cl_2$ ★ chloride ions consumed

$$E_{cal} = E^\circ_{cal} - \frac{0.0591}{2} \log \frac{[Hg_2^{2+}][Cl]^2}{(Hg_2Cl_2)}$$

* The E_{cal} depends only on the concentration of Cl^-
* Here the $[Cl^-]$ remains constant. Hence E_{cal} also remains constant.

2. Ag - AgCl Electrode



Anode :-

$Ag \rightarrow Ag^+ + 1\bar{e}$

$Ag^+ + Cl^- \rightarrow AgCl$

$Ag + Cl^- \rightarrow AgCl + 1\bar{e}$

Cathode :-

$AgCl \rightarrow Ag^+ + Cl^-$

$Ag^+ + 1\bar{e} \rightarrow Ag$

$AgCl + 1\bar{e} \rightarrow Ag + Cl^-$

$$E_{Ag-AgCl} = E^\circ + \frac{0.0591}{1} \log [Cl^-]$$

depends only on $[Cl^-]$ → always constant.

Faraday's I st & II nd Law.

$$\frac{m}{M} = \frac{Q}{nF}$$

$Q = It$

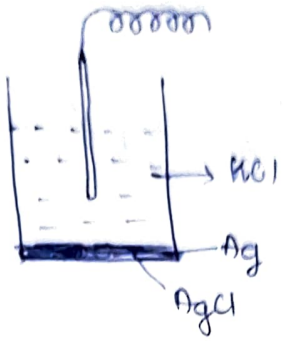
m → mass lib/ deposited

M → molar mass

n → no. of $\bar{e}$

F → Faradays.

2. Ag – AgCl Electrode.



Anode :-

$Ag \rightarrow Ag^+ + 1\bar{e}$

$Ag^+ + Cl^- \rightarrow AgCl$

$Ag + Cl^- \rightarrow AgCl + 1\bar{e}$

Cathode :-

$AgCl \rightarrow Ag^+ + Cl^-$

$Ag^+ + 1\bar{e} \rightarrow Ag$

$AgCl + 1\bar{e} \rightarrow Ag + Cl^-$

$$E_{Ag-AgCl} = E^\circ + \frac{0.0591}{1} \log [Cl^-]$$

depends only on $[Cl^-]$ → always constant.

Faraday's Ist & IInd Law.

$$\frac{m}{M} = \frac{Q}{nF} \quad (Q = It)$$

- m → mass lib/ absorbed deposited
- M → molarmass
- n → no. of $\bar{e}$
- F → Faradays.